WORKED SOLUTION

STEP II 2017, Question 10

Mechanics • Work Done Against Resistance

STEP II 2017, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q10

The key idea

Newton's second law gives the driving force at every point, and the work done by the engine is the integral of that force over distance. With constant acceleration, $v\frac{\mathrm{d}v}{\mathrm{d}x} = a$ turns an integral in x into one in v. The only subtlety is that an engine can push but not pull: when the force needed becomes negative, the brakes do that job and the engine contributes nothing, so we integrate only where the driving force is positive.

Solution

Model the car as a particle moving along the x-axis, with driving force F from the engine. While the car is moving forwards the resistance $Av^2 + R$ acts backwards.

First half. Newton's second law gives $F - (Av^2 + R) = ma$, so

$$F = ma + R + Av^2,$$

which is positive throughout. The work done by a force F moving its point of application through $\mathrm{d}x$ is $F\,\mathrm{d}x$, so the work done by the engine over the first half is

$$\int_0^d (ma + R + Av^2)\,\mathrm{d}x,$$

as required. For the change of variable, the acceleration is $v\frac{\mathrm{d}v}{\mathrm{d}x} = a$, so $\mathrm{d}x = \frac{v}{a}\,\mathrm{d}v$. Since $v^2 = 2ax$ (constant acceleration from rest), v runs from 0 at $x = 0$ to $\sqrt{2ad} = w$ at $x = d$. Hence the work is

$$\int_0^w \frac{(ma + R + Av^2)\,v}{a}\,\mathrm{d}v,$$

as required. Evaluating it, and using $w^2 = 2ad$:

$$W_1 = \frac{1}{a}\left[\frac{(ma+R)v^2}{2} + \frac{Av^4}{4}\right]_0^w = \frac{(ma+R)w^2}{2a} + \frac{Aw^4}{4a} = (ma+R)d + Aad^2.$$

Second half. Now the acceleration is $-a$, so $F-(Av^2+R)=-ma$, giving

$$F = Av^2 + R - ma.$$

The car slows from w to 0, with $v\frac{\mathrm{d}v}{\mathrm{d}x} = -a$, so $\mathrm{d}x = -\frac{v}{a}\,\mathrm{d}v$. Over any stretch where $F>0$ and the speed falls from v_1 to v_2 (with $v_1 > v_2$), the engine does work

$$\int_{v_1}^{v_2} F\left(-\frac{v}{a}\right)\mathrm{d}v = \int_{v_2}^{v_1} \frac{(Av^2+R-ma)\,v}{a}\,\mathrm{d}v. \qquad (*)$$

Part (i)

If $R > ma$ then $F = Av^2 + (R-ma) > 0$ for every v, so the engine works throughout the second half. By $(*)$ with $v_1 = w$, $v_2 = 0$:

$$W_2 = \frac{1}{a}\left[\frac{(R-ma)v^2}{2} + \frac{Av^4}{4}\right]_0^w = (R-ma)d + Aad^2.$$

Adding, the $ma\,d$ terms cancel and the total work is

$$W_1 + W_2 = 2Aad^2 + 2Rd,$$

as required.

Part (ii)

Now $R < ma$, so $F = Av^2 - (ma-R)$ is zero exactly when

$$v = u, \qquad \text{where } u^2 = \frac{ma-R}{A} > 0.$$

This speed is actually reached during the second half if $u < w$, that is, if $\frac{ma-R}{A} < 2ad$, which is precisely the condition $R > ma - 2Aad$. So as the car slows from w to 0, F is positive while $v > u$, falls to zero at the speed u, and is negative (braking) for $v < u$. This proves the first claim.

> **Ask yourself** Why can't we just reuse the answer to (i)?

The integral in (i) adds up $F\,\mathrm{d}x$ over the whole second half. Here F is negative for $v < u$, and that stretch would *subtract* from the total. But the engine isn't doing negative work there. It does nothing, and the brakes take over. So we stop the integral at $v = u$. By $(*)$ with $v_1 = w$, $v_2 = u$:

$$\begin{aligned} W_2 &= \frac{1}{a}\left[\frac{(R-ma)v^2}{2} + \frac{Av^4}{4}\right]_u^w \\ &= (R-ma)d + Aad^2 - \frac{1}{a}\left(\frac{(R-ma)u^2}{2} + \frac{Au^4}{4}\right). \end{aligned}$$

Substituting $Au^2 = ma - R$, the bracket is

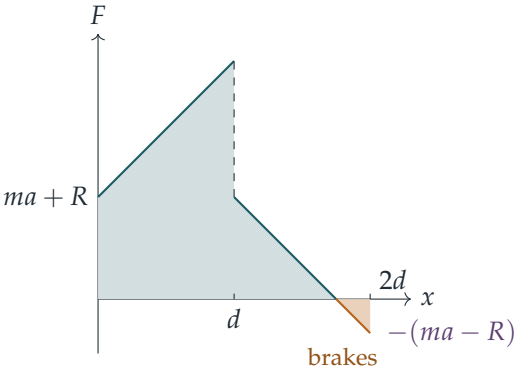
$$-\frac{(ma-R)^2}{2A} + \frac{(ma-R)^2}{4A} = -\frac{(ma-R)^2}{4A},$$

so $W_2 = (R - ma)d + Aad^2 + \frac{(ma-R)^2}{4Aa}$. The first half is unchanged (there $F > 0$ always), so the total is

$$W_1 + W_2 = 2Aad^2 + 2Rd + \frac{(ma - R)^2}{4Aa},$$

as required.

A picture. Since $v^2 = 2ax$ in the first half and $v^2 = 2a(2d - x)$ in the second, F is a *linear* function of x on each half, and the engine's work is the area under the positive part of the graph. In case (ii) the graph dips below the axis near the end. The missing triangle has height $ma - R$ and, since F falls at rate $2Aa$ per unit distance, base $\frac{ma-R}{2Aa}$. Its area $\frac{(ma-R)^2}{4Aa}$ is exactly the extra term.



Checks

- At the boundary $R = ma$ the extra term vanishes, so the answers to (i) and (ii) agree, as they must, since then F only reaches zero at the very end.
- At the other boundary $R = ma - 2Aad$, we have $u = w$, the engine does nothing in the second half, and the total should be $W_1 = (ma + R)d + Aad^2$. Substituting $ma - R = 2Aad$ into the answer to (ii) gives $2Aad^2 + 2Rd + Aad^2 = 2Rd + 3Aad^2$, and $(ma + R)d + Aad^2 = 2Rd + 2Aad^2 + Aad^2$ also, so they agree.
- Numerically, with $m = a = 1$, $A = \frac{1}{2}$, $d = 2$, $R = \frac{1}{2}$ (the case drawn above), the formula in (ii) gives $4 + 2 + \frac{1}{8} = 6.125$, and a direct numerical integration of the positive part of F over the journey gives the same value.
- Units: Aad^2, Rd and $(ma - R)^2/(Aa)$ are all a force times a distance, so they are all energies.

Common slips

- Forgetting that $\mathrm{d}x = -\frac{v}{a}\,\mathrm{d}v$ in the second half, which reverses the limits; a sign error here makes the second-half work negative.

- Integrating the negative part of F in (ii), which just reproduces the answer to (i).
- Not checking that the speed u is actually reached before the car stops: that is what the lower bound on R guarantees.

Beyond STEP

The work done by a force is a line integral, $\int \mathbf{F} \cdot \mathrm{d}\mathbf{r}$, and the change of variable used here is the one-dimensional version of the work–energy theorem. Resistances like Av^2 are not conservative, so the work depends on how the journey is driven, not just where it starts and ends. The engine's "one-sided" behaviour is an early taste of the constraints that appear in optimal control problems, such as finding the most fuel-efficient way to drive a given distance. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats work, energy and velocity-dependent forces carefully.

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